

Semantic properties and combinatorial restrictions of attitude predicates

Day 1: Introduction

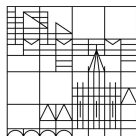
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Konstanz



Preliminaries

Our empirical domain

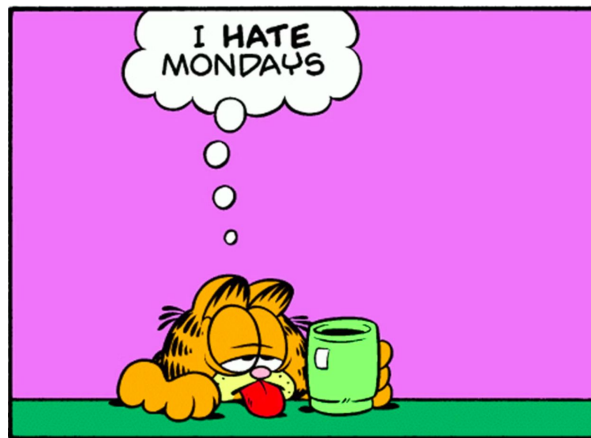
Attitude predicates

- combine with **clauses**
- and ascribe to an agent an attitude

- (1) a. Alice **believes** **that** Noël is happy.
b. Alice **is curious** **whether** Noël is happy.

verbs or adjectives

declarative, interrogative, ...
beliefs, desires, speech, ...



Our empirical domain

These are actively studied by

- philosophers,
- linguists,
- psychologists,
- logicians

who use a broad range of methodologies:

- theoretical,
- experimental,
- computational

B

JO HEARD THAT IT WAS RAINY OUTSIDE.



CAN YOU USE THE FOLLOWING SENTENCE IN THIS CONTEXT?

AND SHE LEMS THAT IT'S RAINY OUTSIDE.

YES

NO

Maldonado et al. (2022)

Three categories of attitude predicates

Attitude predicates differ in their combinatorial properties,
= the clause types that they combine with.

Responsives

✓ declaratives, ✓ interrogatives

(2) Cleo knows {that, whether} Petra is happy.

Anti-rogatives

✓ declaratives, ✗ interrogatives

(3) Cleo believes {that, *whether} Petra is happy.

Rogatives

✗ declaratives, ✓ interrogatives

(4) Cleo wonders {*that, whether} Petra is happy.

A syntactic question, a semantic answer

- | | |
|---|-----------------|
| (2) a. Cleo knows {that, whether} Petra is happy. | [responsive] |
| b. Cleo believes {that, *whether} Petra is happy. | [anti-rogative] |
| c. Cleo wonders {*that, whether} Petra is happy. | [rogative] |

The question

What explains the combinatorial properties of attitude predicates?

The answer strategy

- These predicates mean different things (=have different semantic properties).
- Let's use these semantic properties to explain their combinatorial properties.

What if there is nothing to explain?

- | | |
|---|-----------------|
| (2) a. Cleo knows {that, whether} Petra is happy. | [responsive] |
| b. Cleo believes {that, *whether} Petra is happy. | [anti-rogative] |
| c. Cleo wonders {*that, whether} Petra is happy. | [rogative] |

A possibility

These patterns are accidental, underivable facts about these verbs.

- “Believe” selects for declaratives and not interrogatives.
- “Wonder” selects for interrogatives and not declaratives.
- These facts have to be memorized on a case by case basis.
 - ~ “hit” is transitive, “jump” is intransitive...

There is something to explain (1)

- | | |
|---|-----------------|
| (2) a. Cleo remembers {that, whether} Petra is happy. | [responsive] |
| b. Cleo supposes {that, *whether} Petra is happy. | [anti-rogative] |
| c. Cleo is curious {*that, whether} Petra is happy. | [rogative] |

Similar meanings, similar combinatorial restrictions

- within the same language,

There is something to explain (1)

- | | |
|---|-----------------|
| (2) a. Cleo sait {que, si} Petra est heureuse. | [responsive] |
| b. Cleo croit {que, *si} Petra est heureuse. | [anti-rogative] |
| c. Cleo se demande {*que, si} Petra est heureuse. | [rogative] |

Similar meanings, similar combinatorial restrictions

- within the same language,
- across languages.

There is something to explain (2)

- (3) a. ??Cleo is sure whether Petra is happy. [also: care, be certain]
b. Cleo isn't sure whether Petra is happy.
- (4) a. *Dani believes which cyclist won. [also: imagine, think]
b. Dani can't believe which cyclist won.

Combinatorial properties sensitive to environment

- Negation, modals, etc., sometimes affect these properties.
- Selection is a local relationship, this is unexpected.
~ “jump” is intransitive, but becomes transitive when negated.
- Why memorize... when you can (try to) derive!

The path forward

- The combinatorial properties of attitude predicates aren't accidental facts.
(in the general case)
- They depend on the semantic properties of those predicates,
- and on those of the environment they are in, like under negation.

Given this...

Let's use predicates' semantic properties to explain their combinatorial properties!

After establishing them as precisely as possible, and cross-linguistically.

By the end of the week you will...

- know a lot about the contemporary findings and debates on attitude reports
- know a lot about various methodologies used in linguistic analysis
- have adopted an attitude predicate and maybe a research topic of your own

Housekeeping

The MECORE project

MECORE: A cross-linguistic investigation of **ME**aning-driven **CO**mbinatorial **RE**strictions in clausal embedding

- Develops diagnostics for ~20 semantic properties and ~15 selectional properties
- Samples ~50 predicates and annotate their properties for ~14 languages.
- Evaluates/discovers cross-ling generalisations about semantics-selection correlations.



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If you want to get involved this week/in general

- Course materials and more at: deniz.fr/esslli
- Please fill out our [let us get to know you form](#)
This is to gauge what languages & academic backgrounds are represented.
- There is [a mini project](#) (for Aug. 4th):
 - Adopt an attitude predicate and describe it
 - See if it abides by or challenges what we present
 - Results to be discussed on the last day
- Explore the [MECORE](#) or the [MegaAttitude](#) databases.

Major generalizations about attitude predicates

A necessary and sufficient condition for responsiveness?

Veridicality and responsiveness

(Egré 2008)

A predicate is responsive iff it is veridical.

Veridicality

A predicate V is veridical iff “ x Vs p ” entails p

- | | | | | |
|-----|--------------------------------------|----------------|---------------|---------------|
| (5) | a. Alice knows that it's raining. | $\Rightarrow$ | It's raining. | veridical |
| | b. Alice believes that it's raining. | $\nRightarrow$ | It's raining. | non-veridical |

A necessary and sufficient condition for responsiveness?

Veridicality and responsiveness

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A predicate is responsive iff it is veridical.

Verifying cases

- If a verb is veridical, then it is responsive. (if)
(6) Alice knows whether it's raining.

- If a verb is not veridical, then it is anti-rogative. (only if)
(7) a. *Alice believes whether it's raining.
b. *Alice believes which cyclist won.

A necessary and sufficient condition for responsiveness?

Veridicality and responsiveness

(Egré 2008)

A predicate is responsive iff it is veridical.

Counterexamples to “if a verb is veridical, then it is responsive.”

Few and self-contained.

- (8) a. Alice is right that Tadej won. $\Rightarrow$ Tadej won.
b. *Alice is right whether Tadej won.

“Be right” is veridical, but not responsive. Also: “regret” and “it is true that.”

A necessary and sufficient condition for responsiveness?

Veridicality and responsivity

(Egré 2008)

A predicate is responsive iff it is veridical.

Counterexamples to “If a verb is not veridical, then it is anti-rogative.”

More difficult to sweep under the rug.

- (9) a. Alice decided that Tadej won. $\nRightarrow$ Tadej won.
 b. Alice decided whether Tadej won.

“Decide” is non-veridical & responsive. Also: “doubt,” “be certain,” “agree,” “tell,” ...

(Egré’s paper is all about anticipating, and defusing these counter-examples.)

In search for sufficient conditions

The class of non-veridicals contains

- responsiveness, and
- anti-rogatives

We'd like to know what makes a predicate belong to either category.

So in come additional generalizations.

Note: Don't worry about memorizing all of these notions.

A sufficient condition for responsivity?

Change of state and responsivity (White & Rawlins 2019)

If an attitude predicate is change of state, it is responsive.

(Covers veridicals and non-veridicals alike.)

Verifying cases

(10) a. Alice decided {to, whether to} leave.

b. Alice intended {to, *whether to} leave. (not strictly a prediction)

Not in focus this week, but maybe related to events & attitudes on day 3:

(11) Alice {#thinks, was thinking} whether she should leave.

A sufficient condition for anti-rogativity: Neg-raising?

Neg-raising and anti-rogativity

If a predicate is neg-raising, it is anti-rogative.

(Zuber 1982, Egré 2008, Mayr 2019, Theiler et al. 2019)

Neg-raising

A predicate V is neg-raising iff “ x doesn’t V p ” entails “ x V s not p ”

(12) Alice doesn’t believe that it’s raining. [neg-raising]

⇒ Alice believes that it’s not raining.

(13) Alice doesn’t claim that it’s raining. [non-neg-raising]

⇏ Alice claims that it’s not raining.

A sufficient condition for anti-rogativity: Neg-raising?

Neg-raising and anti-rogativity

If a predicate is neg-raising, it is anti-rogative.

Verifying cases

- (14) a. *Alice believes whether it's raining.
b. *Alice thinks whether it's raining.

A sufficient condition for anti-rogativity: Neg-raising?

Neg-raising and anti-rogativity

If a predicate is neg-raising, it is anti-rogative.

Potential counterexamples (Roberts 2021, White 2021, Özyıldız 2021)

- (15) a. Ursula **can't believe which** candidate won the HOA election.
b. We **can choose to believe whether** the word of God is true or not.
c. I'm **trying to think whether** I'd have been a star today or not.

A sufficient condition for anti-rogativity: Preferentiality?

Preferentiality and anti-rogativity

(Uegaki & Sudo 2019)

If a predicate is non-veridical and preferential, it is anti-rogative.

Preferential predicates

- Non-veridical: hope, fear, wish, want, prefer...
- Veridical: be surprised, be annoyed, be glad, like, hate, ...

A sufficient condition for anti-rogativity: Preferentiality?

Preferentiality and anti-rogativity

(Uegaki & Sudo 2019)

If a predicate is non-veridical and preferential, it is anti-rogative.

Verifying cases

- (16) a. *Ben hopes/wishes which students you invited.
b. Ben was surprised/likes which students you invited.

A sufficient condition for anti-rogativity: Preferentiality?

Preferentiality and anti-rogativity

(Uegaki & Sudo 2019)

If a predicate is non-veridical and preferential, it is anti-rogative.

Potential counterexamples

(White 2021, MECORE team 2022)

(17) a. I was hoping whether you'd be able to guide me.

b. I fear whether I'll have use of my arms by the age of 60.

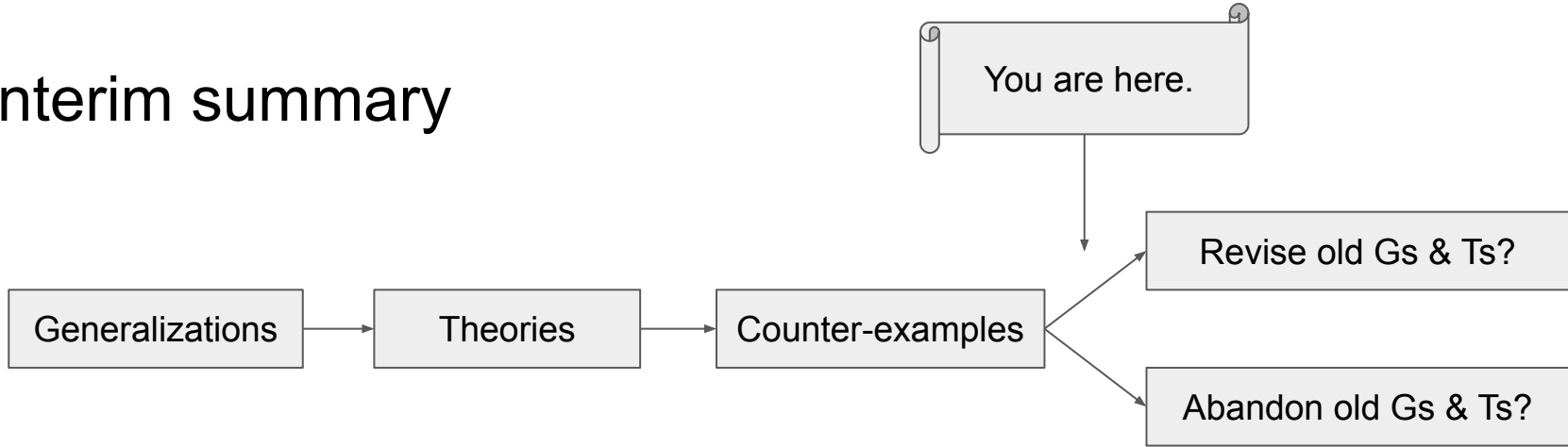
Generalizations on *subtypes* of clauses

- Declaratives and interrogatives have subtypes.
And some generalizations govern *subtypes* of clauses.
 - Subtypes of (embedded) interrogatives:
 - Polar: ...whether Tadej won.
 - Alternative: ...whether Tadej or Jonas won.
 - Constituent: ...which cyclist won.
 - **Karttunen's (1977) generalization:**
Emotive factives are incompatible with *whether* interrogatives.
- (18) a. Alice was surprised which cyclist won.
b. #Alice was surprised whether Tadej (or Jonas) won.

Generalizations on *subtypes* of clauses

- We lumped declaratives and interrogatives together, but these have subtypes. And some generalizations govern *subtypes* of clauses.
 - Subtypes of (embedded) declaratives in Spanish:
 - Indicative
 - Subjunctive
 - **Villalta's (2008) generalization:**
A predicate can take subjunctive complements iff it is focus sensitive.
- (19) I want Nadine to teach on TUESDAY ≠ I want NADINE to teach on Tuesday.
Querer (=want) is focus sensitive, and takes subjunctive complements.

Interim summary



General expectations

- If veridical, then responsive.
- Else if
 - Change of state, then responsive.
 - Neg-raising, then anti-rogative.
 - Preferential, then anti-rogative.

Uniform approaches to clausal embedding

From description to explanation

- We've seen generalizations that relate combinatorial and semantic properties.
- *Why* should such generalizations hold?

Before turning to specific accounts, we need a general framework.

One non-uniform view on clausal embedding

- Embedded declaratives denote sets of possible worlds (= propositions)

$$\begin{aligned}\llbracket \text{that it's raining} \rrbracket &= \{w : \text{it's raining at } w\} \\ &= \lambda w.\text{rain}(w)\end{aligned}$$

- Embedded interrogatives denote sets of propositions
These propositions are *answers* to the interrogative

$$\begin{aligned}\llbracket \text{whether it's raining} \rrbracket &= \{\lambda w.\text{rain}(w), \lambda w.\sim\text{rain}(w)\} \\ \llbracket \text{who is Dutch} \rrbracket &= \{\lambda w.\text{dutch}(a,w), \lambda w.\text{dutch}(b,w), \dots\}\end{aligned}$$

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These propositions are *answers* to the interrogative

$\llbracket \text{whether it's raining} \rrbracket = \{\lambda w.\text{rain}(w), \lambda w.\sim\text{rain}(w)\}$

$\llbracket \text{who is Dutch} \rrbracket = \{\lambda w.\text{dutch}(a,w), \lambda w.\text{dutch}(b,w), \dots\}$

- Embedded declaratives denote sets of propositions as well

$\llbracket \text{that it's raining} \rrbracket = \{\lambda w.\text{rain}(w)\}$

- This is Proposition to Question reduction.
(There are alternatives to, and different ways of doing P-to-Q.)

One uniform view on clausal embedding

Attitude verbs denote relations between (sets of) propositions and individuals.

- $\llbracket \text{believe} \rrbracket = \lambda Q. \lambda x. \exists p \in Q: \text{believe}'(x, p)$
- $\llbracket \text{know} \rrbracket = \lambda Q. \lambda x. \exists p \in Q: p \ \& \ \text{believe}'(x, p)$
- $\llbracket \text{wonder} \rrbracket = \lambda Q. \lambda x. \text{wonder}'(x, Q)$

Sample semantic derivation (1)

(1) $\llbracket \text{Alice} \rrbracket = a$
 $\llbracket \text{believe} \rrbracket = \lambda Q. \lambda x. \exists p \in Q: \text{believe}'(x, p)$
 $\llbracket \text{that it's raining} \rrbracket = \{\text{rain}\}$

(2) $\llbracket \text{Alice believes that it's raining} \rrbracket$
 $= \llbracket \text{believe} \rrbracket (\llbracket \text{that it's raining} \rrbracket) (\llbracket \text{Alice} \rrbracket)$
 $= [\lambda Q. \lambda x. \exists p \in Q: \text{believe}'(x, p)](\{\text{rain}\})(a)$
 $= \exists p \in \{\text{rain}\}: \text{believe}'(a, p)$
 $= \text{believe}'(a, \text{rain})$

Uh oh...?

If clauses uniformly denote sets of propositions,
and attitude verbs indiscriminately take sets of propositions...

- “know” should be compatible with declaratives and interrogatives ✓
- “believe” should be compatible with declaratives and interrogatives ✗
- “wonder” should be compatible with declaratives and interrogatives ✗

Sample semantic derivation (2)

$$\begin{aligned}(3) \quad & \llbracket \text{Alice} \rrbracket = a \\ & \llbracket \text{believe} \rrbracket = \lambda Q. \lambda x. \exists p \in Q: \text{believe}'(x, p) \\ & \llbracket \text{whether it's raining} \rrbracket = \{\text{rain}, \sim \text{rain}\}\end{aligned}$$

$$\begin{aligned}(4) \quad & \llbracket \text{Alice believes whether it's raining} \rrbracket \\ &= \llbracket \text{believe} \rrbracket (\llbracket \text{whether it's raining} \rrbracket) (\llbracket \text{Alice} \rrbracket) \\ &= [\lambda Q. \lambda x. \exists p \in Q: \text{believe}'(x, p)] (\{\text{rain}, \sim \text{rain}\}) (a) \\ &= \exists p \in \{\text{rain}, \sim \text{rain}\}: \text{believe}'(a, p)\end{aligned}$$

!!! We can derive truth conditions for ungrammatical attitude reports...

Truth conditions of ungrammatical sentences?

We can derive truth conditions for ungrammatical attitude reports.

This looks like a wrong prediction, but it's exactly what we want.

If it's meanings that derive combinatorial restrictions... we should know what those meanings are.

*What bothers us is WHY, e.g., we have “believe that” but not “*believe who,” even though the latter looks as if it ought to make sense (cf. “have an opinion as to who”).*

Boër (1978: p. 312)

Meaning driven unacceptability in general

- Some unacceptable sentences are not unacceptable for syntactic reasons, but because they have deviant meanings.
- Usually, “deviant” means “tautologous” or “contradictory.”
- This kind of deviance is perceived as unacceptability. (Gajewski 2005)

- (5) a. *I don't know what a fool Bill is.
b. *There is every dog in the room.
c. *Samson bit anyone.
d. *How many kids don't you have?

Grimshaw (1979)

Barwise & Cooper (1981)

Chierchia (2006)

Fox & Hackl (2006)

Meaning driven unacceptability for “believe” and “wonder”

The same idea extends to sentences like (1) and (2).

- We may freely put together “believe” and “whether p,” “wonder” and “that p.”
- But, given independent assumptions about the meanings of these pieces,
we derive tautologies or contradictions...

(7) *Alice believes whether it's raining.

The meaning of “believe” + neg-raising + the meaning of “whether” = $\perp$ / $\top$

(8) *Alice wonders that it's raining.

The meaning of “wonder” + the meaning of “that” = $\perp$

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The same idea extends to sentences like (1) and (2).

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Next section!

The meaning of “believe” + neg-raising + the meaning of “whether” = $\perp$ / $\top$

(8) *Alice wonders that it’s raining.

Ciardelli et al. (2019)

The meaning of “wonder” + the meaning of “that” = $\perp$

Two questions we won't directly address, but ask us!

- Is there an asymmetry between accounting for “If X, then responsive” vs. “If X, then anti-rogative”?
- Is there any way of *saving* a sentence after a tautology or a contradiction has been derived, or *preventing* one from arising?

See Roberts (2021) on “can’t believe” and Mayr (2019) on “not sure.”

A case study: Zuber 1982

Know vs believe/think/seem

- | | |
|--|-----------------|
| (1) Cleo knows {that, whether} Petra is happy. | [responsive] |
| (2) Cleo believes {that, *whether} Petra is happy. | [anti-rogative] |
| (3) Cleo thinks {that, *whether} Petra is happy. | [anti-rogative] |
| (4) It seems {that, *whether} Petra is happy. | [anti-rogative] |

Puzzle: why can't predicates such as *believe*, *think*, and *seem* take questions?

Zuber (1982): They can't because these predicates are **neg-raising**

Neg-raising

V is neg-raising (neg-transportation): “not V that S” implies “V that not S”

Canonical neg-raisers: *believe, think, seem, be likely, want*, etc.

- (1) Bill does not believe that Sue will call.
- (2) Bill believes that Sue will not call.

Neg-raising

V is neg-raising (neg-transportation): “not V that S” implies “V that not S”

Canonical neg-raisers: *believe, think, seem, be likely, want*, etc.

- (1) Bill does not believe that Sue will call.
- (2) Bill believes that Sue will not call.

- (3) It does not seem that it will rain.
- (4) It seems that it will not rain.

Examples of non-neg-raising predicates

V is neg-raising (neg-transportation): “not V that S” implies “V that not S”

Know and *say* are not neg-raising:

- (1) Bill does not know that Sue will call.
- (2) Bill knows that Sue will not call.
 - (1) does not imply (2),

Examples of non-neg-raising predicates

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Know and *say* are not neg-raising:

- (1) Bill does not know that Sue will call.
- (2) Bill knows that Sue will not call.
 - (1) does not imply (2),

- (3) Bill did not say that it would rain.
- (4) Bill said that it would not rain.
 - (3) does not imply (4)

What if neg-raisers were to take questions?

(0) “Not V that S” implies “V that not S” [neg-raising]

If a neg-raiser V were able to take questions like *know*, what semantic properties would it have?

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Zuber (1982) postulates the following two additional properties:

(1) “V that not S” implies “V whether S or not”

Cf. “Bill knows that it is not raining” implies “Bill knows whether it is raining.”

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(1) “V that not S” implies “V whether S or not”

Cf. “Bill knows that it is not raining” implies “Bill knows whether it is raining.”

(2) “Not V that S” implies “not V whether S or not”

Cf. “Bill does not know that it is raining” implies “Bill does not know whether it is raining.”

Incompatibility between neg-raising and Zuber's properties

Zuber points out that neg-raising (0) is incompatible with properties (1-2)

- (0) “Not V that S” implies “V that not S” [neg-raising]
- (1) “V that not S” implies “V whether S or not”
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According to (0) and (1), “not V that S” implies “V whether S or not”

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- (0) “Not V that S” implies “V that not S” [neg-raising]
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According to (0) and (1), “not V that S” implies “V whether S or not”

According to (2), “not V that S” implies “not V whether S or not”

This means that “not V that S” will lead to a contradiction.

Recap: Zuber's derivation of **think whether*

Premise (a): *think* is neg-raising, i.e., it satisfies (0)

(0) “Not V that S” implies “V that not S” [neg-raising]

Recap: Zuber's derivation of **think whether*

Premise (a): *think* is neg-raising, i.e., it satisfies (0)

(0) “Not V that S” implies “V that not S” [neg-raising]

Premise (b): if *think* were able to take a *whether*-question, it would satisfy the following two additional properties:

(1) “V that not S” implies “V whether S or not”

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Given (a) and (b), if *think* were able to take a *whether*-question, “not think that S” will lead to a contradiction (as explained in the previous slide), contrary to fact.

Recap: Zuber's derivation of **think whether*

Premise (a): *think* is neg-raising, i.e., it satisfies (0)

(0) “Not V that S” implies “V that not S” [neg-raising]

Premise (b): if *think* were able to take a *whether*-question, it would satisfy the following two additional properties:

(1) “V that not S” implies “V whether S or not”

(2) “Not V that S” implies “not V whether S or not”

Given (a) and (b), if *think* were able to take a *whether*-question, “not think that S” will lead to a contradiction (as explained in the previous slide), contrary to fact.

Therefore, *think* (and neg-raisers in general) cannot take *whether*-questions

Evaluating Zuber's semantic explanation

Empirical: Does Zuber's generalization hold for all neg-raising predicates?

Theoretical: Does Zuber's derivation provide a satisfactory explanation?

- In particular, how reasonable are its premises?

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The two questions are largely intertwined

- The theoretical derivation can often qualify the empirical generalization: apparent counterexamples might not be real counterexamples

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Empirical: Does Zuber's generalization hold for all neg-raising predicates?

Theoretical: Does Zuber's derivation provide a satisfactory explanation?

- In particular, how reasonable are its premises?

The two questions are largely intertwined

- The theoretical derivation can often qualify the empirical generalization: apparent counterexamples might not be real counterexamples
- It is important to check whether the assumptions made in the theoretical derivations are realistic

Zuber's (1982) concluding remark

'It seems important to point out in conclusion that **what I have shown is not that whether-complementizers are absolutely impossible with neg-transportation verbs.** I have shown that they are impossible only with their "main" meaning, the one given by the definition of neg-transportation verb and by the properties Prop 1 and Prop 2. In particular they must be related to the *that*-complementizer in the way shown above. **Consequently it may happen that in some languages one can find, in some construction, *to think whether* or its equivalent. My claim is however that in this case the meaning of the verb will be different from the one stipulated by the properties here discussed.'** (p. 436)

- More on this later in the week

Theoretical evaluation of Zuber's derivation

Theoretical: Does Zuber's derivation provide a satisfactory explanation?

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Premise (b): if *think* were able to take a *whether*-question, it would satisfy the following two additional properties:

- (1) “V that not S” implies “V whether S or not”
- (2) “Not V that S” implies “not V whether S or not”

Zuber's Prop 1

Premise (b-1): if *think* were able to take a *whether*-question, it would satisfy (1)

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(1) “V that not S” implies “V whether S or not”

- “Bill knows that it is not raining” implies “Bill knows whether it is raining or not”
- “Bill said that it was not raining” implies “Bill said whether it was raining or not”

(cf. “Q: Did Bill say whether it was raining or not? A: Yes.”)

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(cf. “Q: Did Bill say whether it was raining or not? A: Yes.”)

- “Bill is sure that it is not raining” implies “Bill is sure (about) whether it is raining or not”

(cf. “Q: Is Bill sure about whether it is raining or not? A: Yes.”)

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(1) “V that not S” implies “V whether S or not”

(1) is indeed generally considered to hold for all question-embedding predicates (see Roelofsen & Uegaki 2020, Özyıldız et al. 2023 for further discussion)

- Premise (b-1) seems reasonable

Floris Roelofsen and Wataru Uegaki. 2020. Searching for a universal constraint on the denotations of clause-embedding predicates. In *Proceedings of Semantics and Linguistic Theory* 30

Deniz Özyıldız, Ciyang Qing, Floris Roelofsen, Maribel Romero, and Wataru Uegaki. 2023. A Crosslinguistic Database for Combinatorial and Semantic Properties of Attitude Predicates. In *Proceedings of the 5th Workshop on Research in Computational Linguistic Typology and Multilingual NLP (SIGTYP 2023)*.

Zuber's Prop 2

Premise (b-2): if *think* were able to take a *whether*-question, it would satisfy (2):

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Zuber motivates this property using *know*:

- “Bill does not know that it is raining” indeed implies “Bill does not know whether it is raining”

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Zuber motivates this property using *know*:

- “Bill does not know that it is raining” indeed implies “Bill does not know whether it is raining”
- But how representative is *know*? Does this property hold for all question-embedding predicates?

Question-embedding predicates without Zuber's Prop 2

Zuber's Prop 2: "Not V that S" implies "not V whether S or not"

(a) Bill did not say that it was raining. (In fact, he said that it was sunny.)

(a') Bill did not say whether it was raining or not.

- (a) does not imply (a'): Prop 2 does not hold for *say*.

Question-embedding predicates without Zuber's Prop 2

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(a) Bill did not say that it was raining. (In fact, he said that it was sunny.)

(a') Bill did not say whether it was raining or not.

- (a) does not imply (a'): Prop 2 does not hold for *say*.

(b) I am not sure that it is raining. (In fact, I am sure that it is sunny.)

(b') I am not sure whether it is raining or not.

- (b) does not imply (b'): Prop 2 does not hold for *be sure*.

Theoretical evaluation of Zuber's derivation

Zuber (1982): If *think* were able to take a *whether*-question, it would satisfy the following two additional properties:

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While (1) can be plausibly taken to hold for all question-embedding predicates, many question-embedding predicates do not satisfy (2), e.g., *say* and *be sure*.

- Egré (2008): (2) in general only holds for factive predicates (e.g., *know*)

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Therefore, Zuber (1982) does not provide a fully satisfactory explanation.

Summary

Zuber attempts to explain why neg-raising predicates such as *think* cannot take questions by pointing out that neg-raising (0) is incompatible with properties (1-2)

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- Strictly speaking, Zuber does not rule out *think whether* altogether. He only claims that if such constructions were attested, then (1) or (2) would fail.

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- However, the extent to which Zuber’s derivation is explanatory depends on whether (1) and (2) hold in general
- (1) does but (2) does not: Zuber’s derivation is not fully explanatory

General implications

Still, Zuber's (1982) pioneering work contains valuable insights and is a great starting point to illustrate a general strategy behind semantic explanations of combinatorial restrictions.

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General implications

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In order to explain why predicates with property X cannot be combined with a certain type of clauses, we can do the following:

- Examine those predicates that do combine with such clauses, and identify the relevant semantic property Y that such predicates share
- Show how the combination of X and Y would lead to a semantic anomaly (contradiction, tautology, etc.)

General implications on evaluation

When evaluating semantic explanations of combinatorial restrictions

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When evaluating semantic explanations of combinatorial restrictions

- **Empirical evaluation:** remember to check whether the assumed semantic properties are satisfied, especially when facing apparent counterexamples.
- **Theoretical evaluation:** remember to critically assess whether the assumed semantic properties are reasonable and hold in general for predicates without such combinatorial restrictions

Theiler et al.'s (2019) derivation of Zuber's generalization

Premise 1: A neg-raising predicate V has an Excluded Middle (EM) presupposition

- (1) EM presupposition: V that S or V that not S
 - Because of this, “not V that S” implies “V that not S” (i.e., neg-raising)

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However, the at-issue content is already presupposed (semantic anomaly)!

Evaluating Theiler et al.'s (2019) derivation

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- It has non-trivial implications on what counts as a neg-raising predicate (for the purpose of the neg-raising generalization): more on this on Day 4

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- It will help us make sense of some apparent counterexamples (Day 4)

Patrick D. Elliott, Nathan Klinedinst, Yasutada Sudo, and Wataru Uegaki. 2017. Predicates of relevance and theories of question embedding. *Journal of Semantics*

Rest of the week

Day 2: Guest lecture by Wataru Uegaki on non-veridical preferentials such as *hope*

(1) Chris liked that Al and Bo were invited to the party.

$\Rightarrow$ Al and Bo were invited to the party. (veridical)

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Anti-rogrativity of *hope*:

(3) Chris liked which students were invited to the party.

(4) *Chris hoped which students were invited to the party.

Rest of the week

Day 3: More on question-embedding *think*

Zuber (1982): ‘Consequently it may happen that in some languages one can find, in some construction, *to think whether* or its equivalent.’

Rest of the week

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Some apparent counterexamples in English from White (2021):

- (1) And it does cause you to **think whether** or not it makes sense for us to be there.
- (2) I was **thinking whether** there was a way to[...]help more than one person.

Rest of the week

Day 4: Evaluating the neg-raising generalization by examining large-scale experimental data and apparent counterexamples from corpora

- What is the best way of analyzing such data to inform our empirical generalizations and theoretical derivations?

Rest of the week

Day 5: Crosslinguistic investigations

Uegaki & Sudo's (2019): non-veridical preferential predicates are anti-rogative (2)

- (1) Chris liked which students were invited to the party.
- (2) *Chris hoped which students were invited to the party.

Karttunen (1977): Emotive factive predicates can take constituent questions (1)
but not *whether*-questions (3)

- (3) *Chris liked whether Al and Bo were invited to the party.

Thank you!

Questions?